

Supporting Information

Simulating organ biomass variability and carbohydrate distribution in perennial fruit crops: a comparison between the common assimilate pool and phloem carbohydrate transport models

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Supplementary Table S1 The optimised parameter values for carbon allocation based on different combinations of leaf availability treatments

Parameters ¹	mean values	Standard deviation ²	CV
	1.48E-3	3.81E-4	0.26
	4.91E-3	2.55E-3	0.52
	3.21E-4	2.06E-4	0.64
	4.90E-3	2.71E-3	0.55
	2.46E-6	3.39E-6	1.37

¹ is the rate of NSC hydrolysis. is secondary growth rate of the radius. is the Michaelis-constant for carbohydrate unloading for NSC synthesis. is the Michaelis-constant for secondary growth. is the Michaelis-constant for berry growth.

² Large variations were found for the Michaelis constants for different processes. One reason was the limiting factors for carbohydrate accumulation and distribution changes under different carbohydrate availability conditions. For example, the sensitivity of structural root NSC to increased steadily with a reduction in the leaf number per vine, raising from 0.38 to 0.81 from the 100 leaves per vine to no leaves treatment. See the sensitivity analysis part in the results and carbohydrate allocation in the discussion. Another reason was we do not have enough data to quantify those parameters.

Supplementary figures:

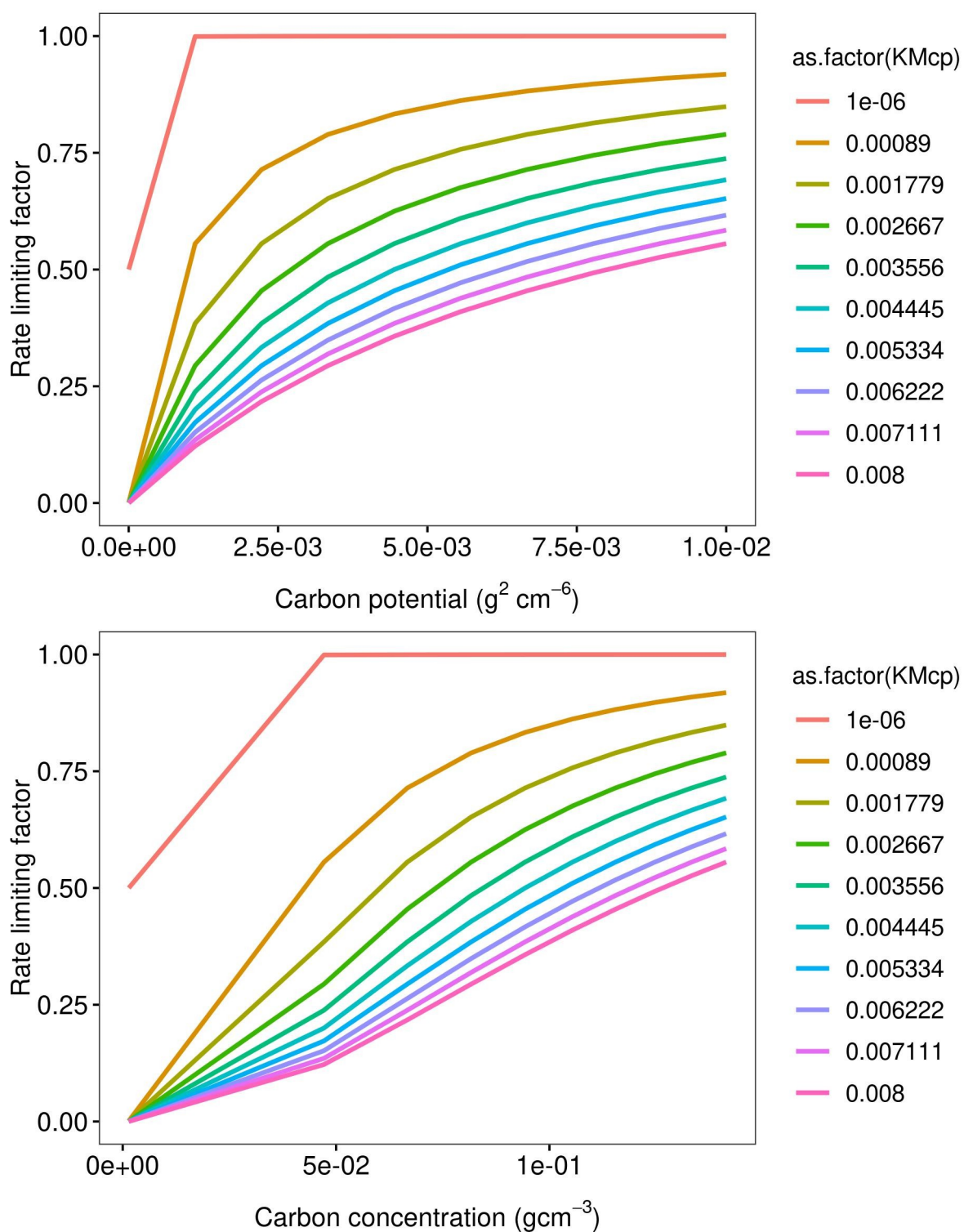


Fig. S1. The effect of K_M values on the rate of sink unloading corresponding to carbon potential (top panel) and carbon concentration (bottom panel).

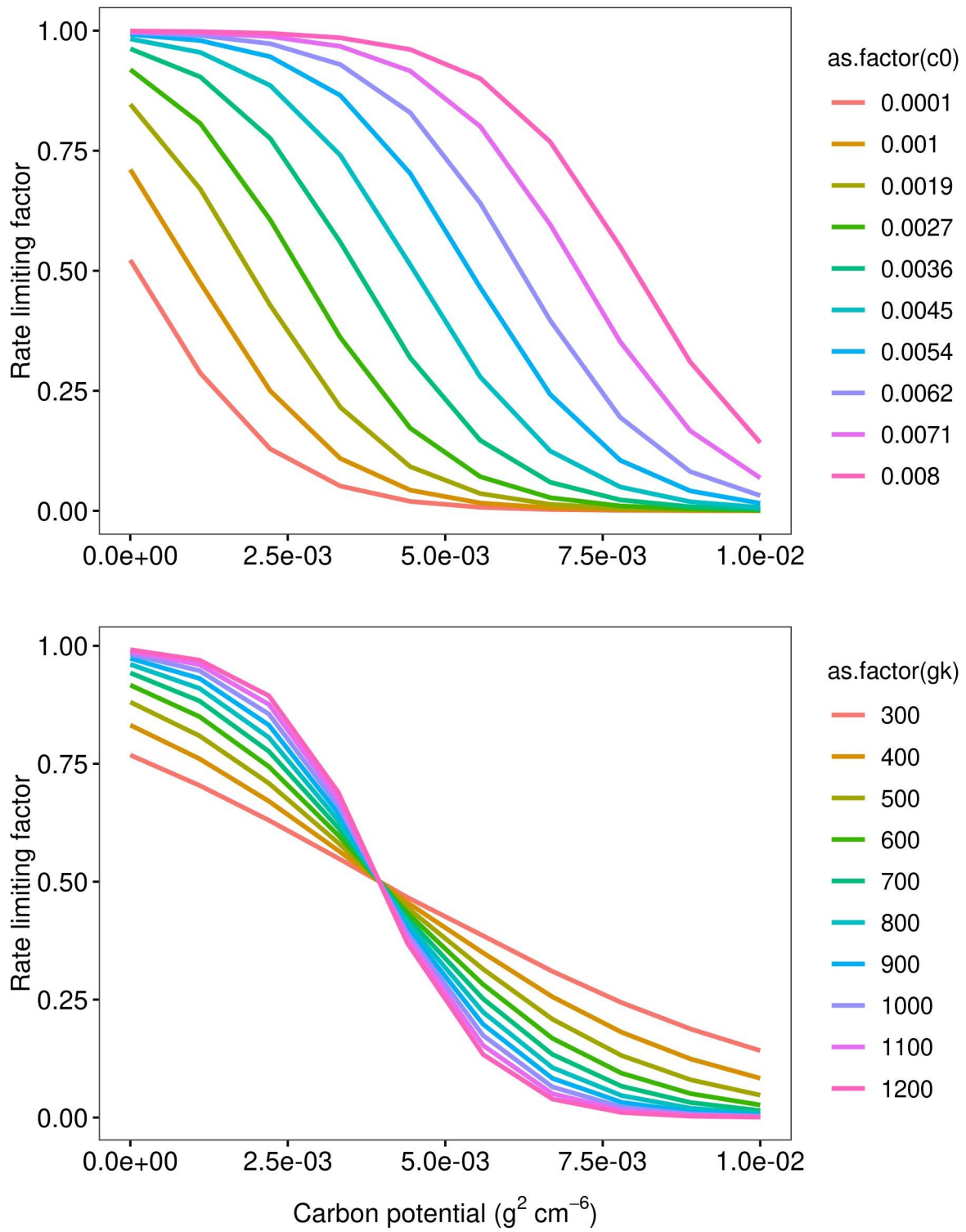


Fig. S2. The effect of inflection point of the source loading function c_0 (top panel) and scaling parameter gk (bottom panel) on the rate of source loading.

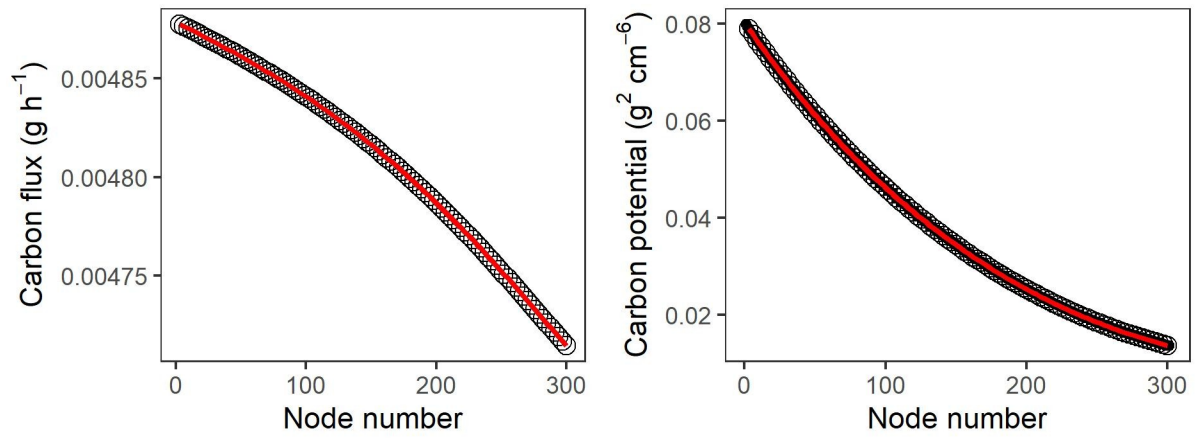


Fig. S3. Comparison of the rates of carbon flux (left panel) and carbon potential (right panel) as outputs from the carbon transport procedure implemented in GroIMP and the original model in L-Studio (Seleznyova and Hanan, 2018), in a single sieve tube which is 30 cm long and 1 cm in radius. The sieve tube was divided into 300 nodes with each internode being 1 cm.



Fig. S4. Illustration of the potted vines that were used in the experiment of Rossouw *et al.* (2017). In total, forty vines were used for the experiment and were distributed in four rows with 10 vines each. The grapevines were enclosed in a bird-proof cage in the hot climate Riverina grape growing region of New South Wales, Australia. The potted vines were spur-pruned to five two-bud spurs in the winter and then trained to have ten shoots per vine where possible.

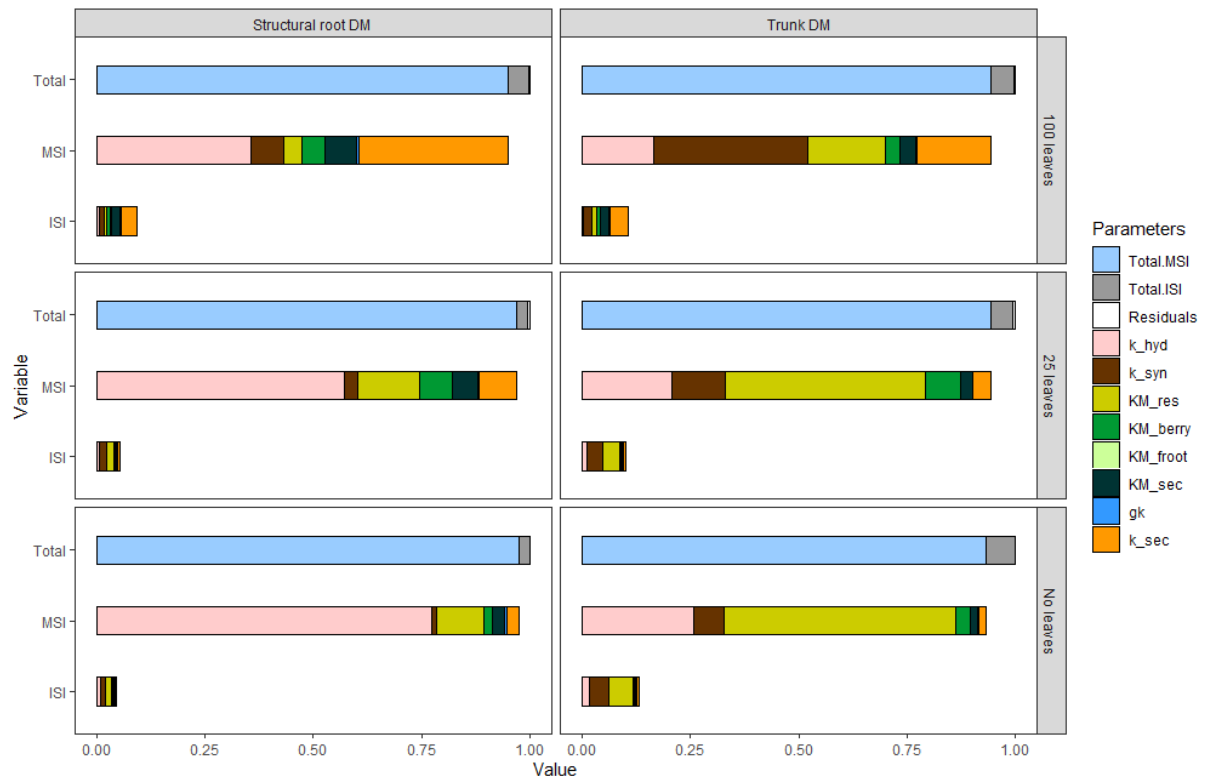


Fig. S5. Global sensitivity analysis of structural root dry matter and trunk dry matter to eight parameters under 100 leaves per vine, 25 leaves per vine and no leaves treatments. The upper bar in each panel shows the distribution of main sensitivity index (*MSI*) and interaction sensitivity index (*ISI*), while the two lower bars show the contribution of each parameter.

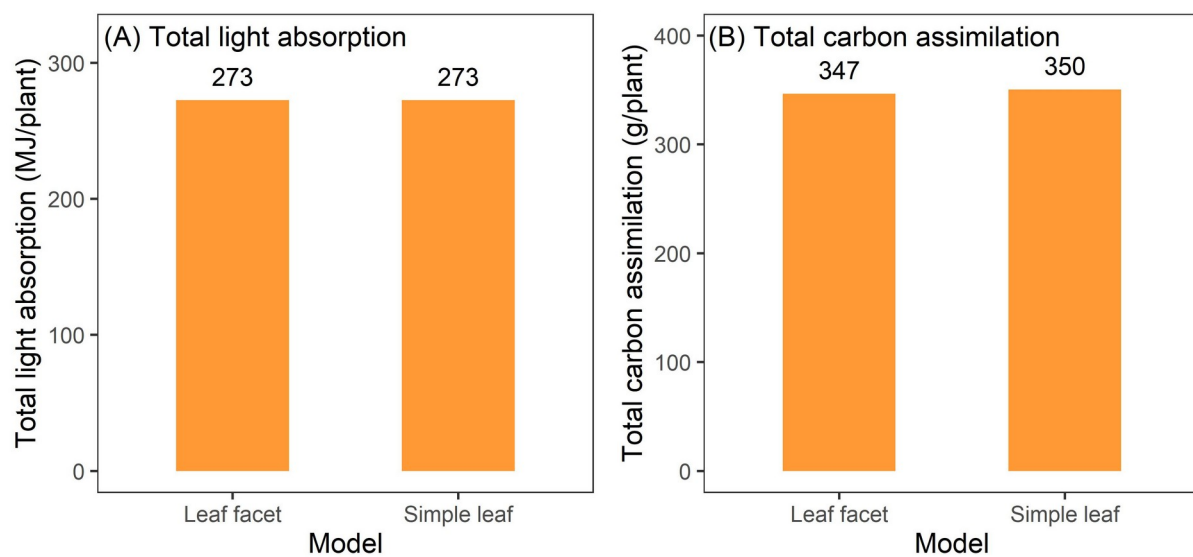


Fig. S6. The effects of leaf angle optimisation method on the total light absorption (a) and carbon assimilation (b) over the 38 simulated days.

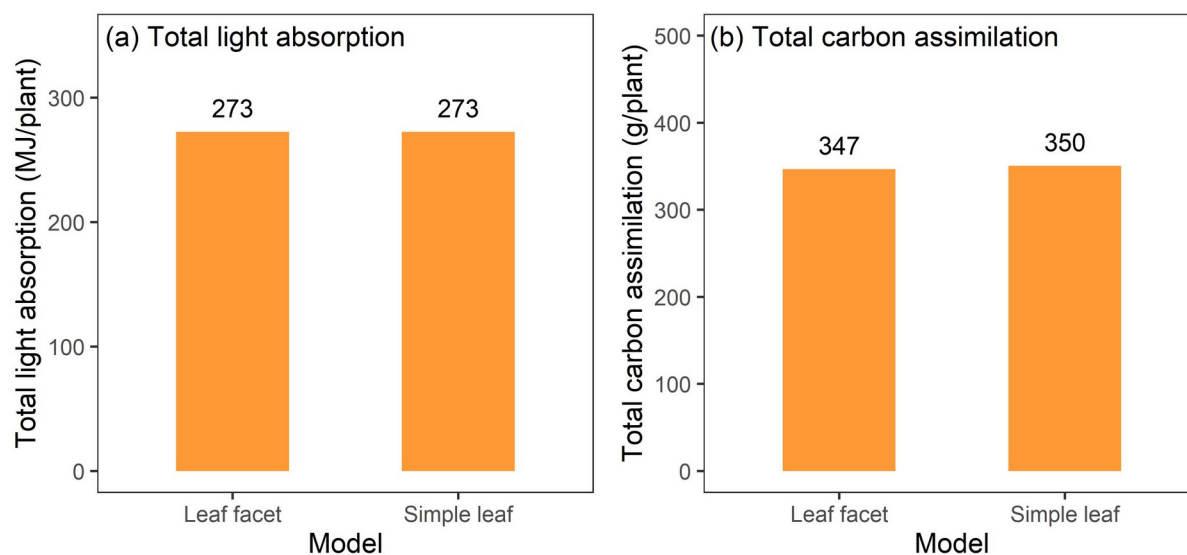


Fig. S7. The effects of leaf facet method on the total light absorption (A) and carbon assimilation (B) over the 38 simulated days. The leaf facet method divided a grapevine leaf into 16 triangles, while the simple leaf method considers a leaf as one facet.

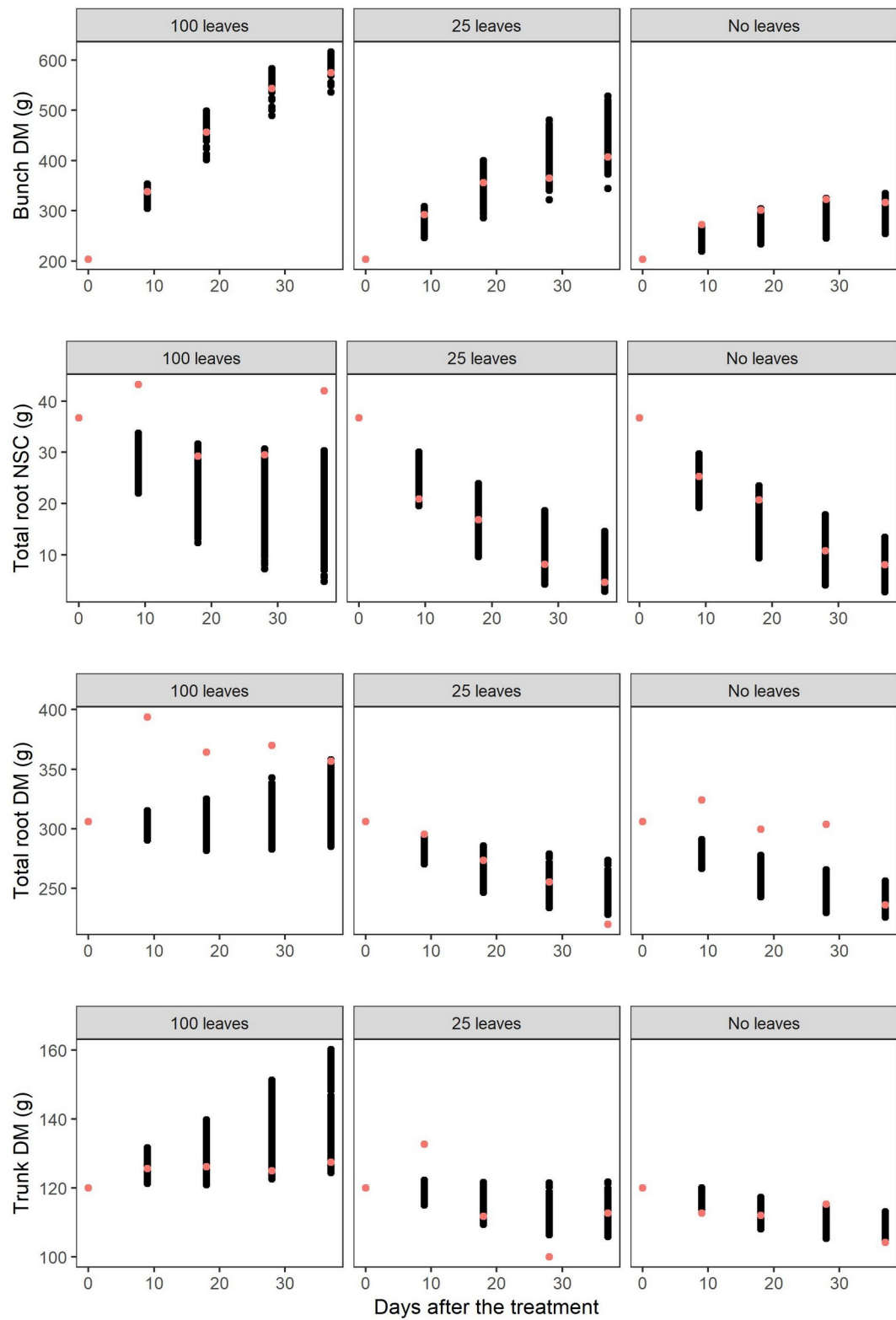


Fig. S8 Distribution of the simulated bunch dry matter (DM) and total root non-structural carbohydrate, total root DM, and trunk DM in the 243 simulations for parameter optimisation in the second round. Simulations were done with different combinations of the values of five parameters: NSC hydrolysis (), secondary growth rate (), the Michaelis constant of NSC synthesis (), and secondary growth (), berry ().

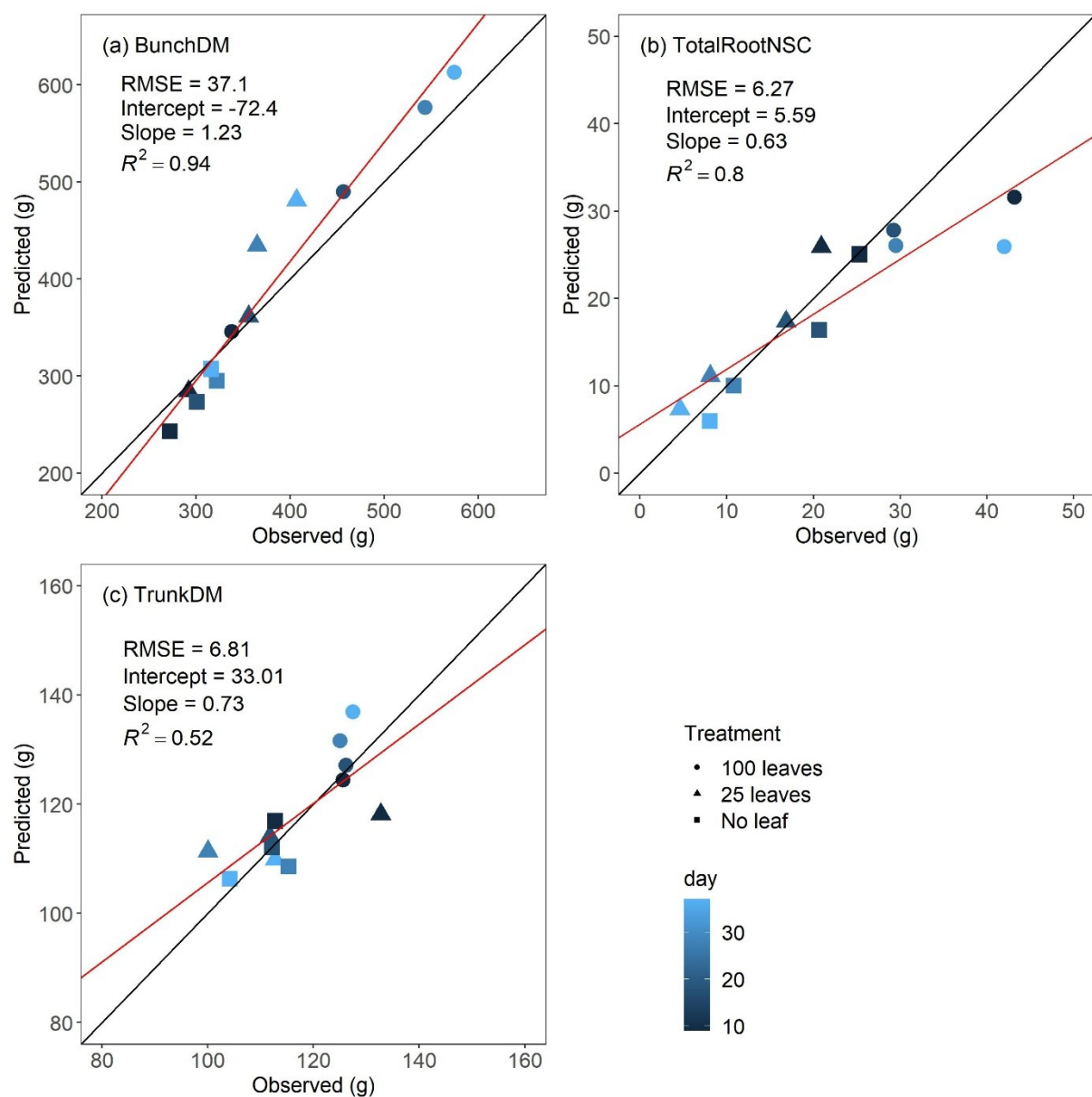


Fig. S9. Verification of the simulated bunch dry matter (a), total root non-structural carbon (b), and trunk dry matter (c) by the GPfit emulator. Parameter values were the optimised means based on different combinations of leaf treatments as shown in Supplementary Table S1.

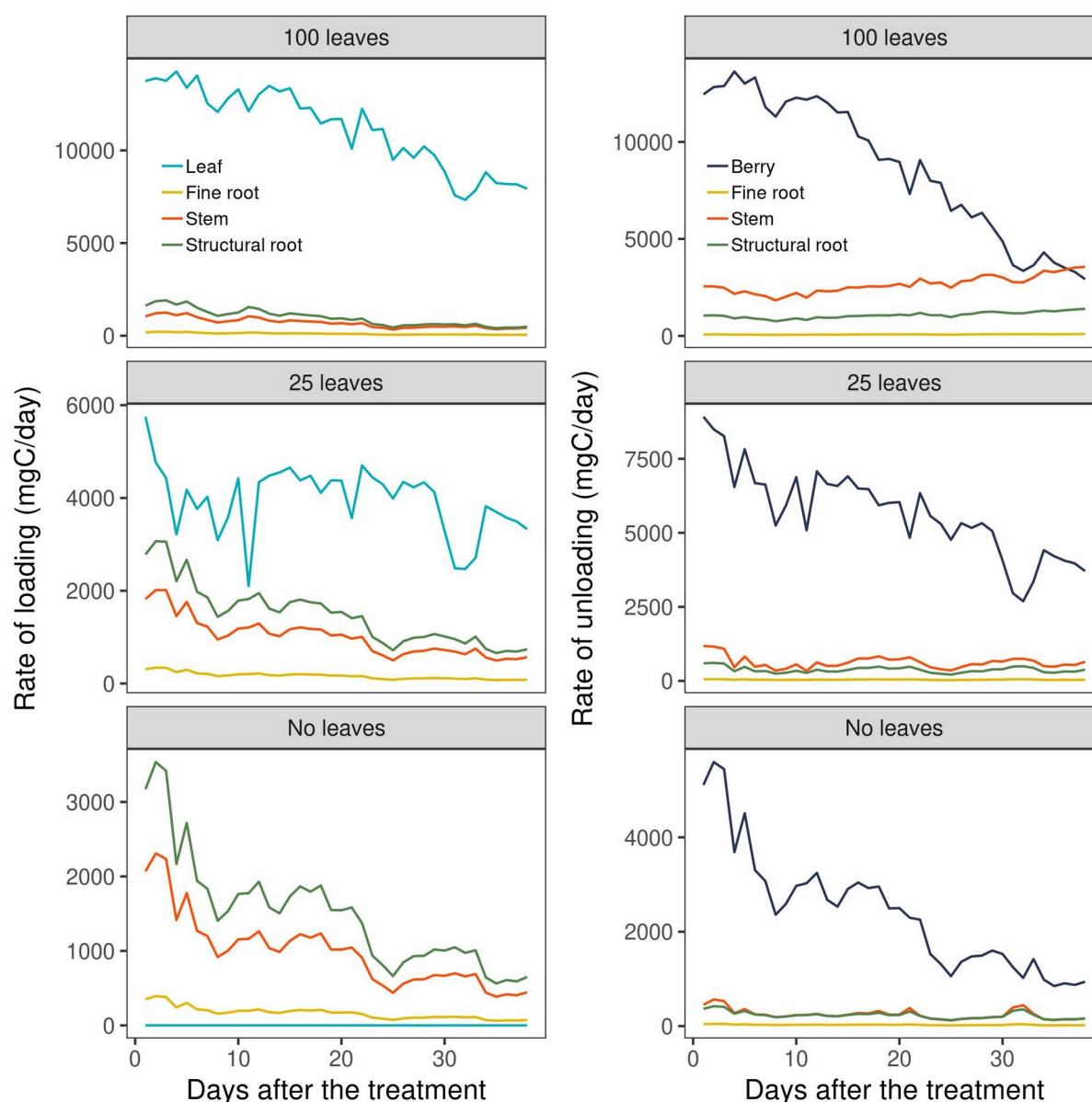


Fig. S10. Simulated daily rates of carbon loading into the phloem by leaves, fine roots, stems and structural roots under the different leaf treatments (left panels) and carbon unloading by berries, fine roots, stems and structural roots (right panels). Stem was a simplified notation for all internodes (the current season's shoot), cordons (2-year-old shoot) and the trunk (the perennial woody part). Daily rates were the sum of the hourly rates of carbon loading.

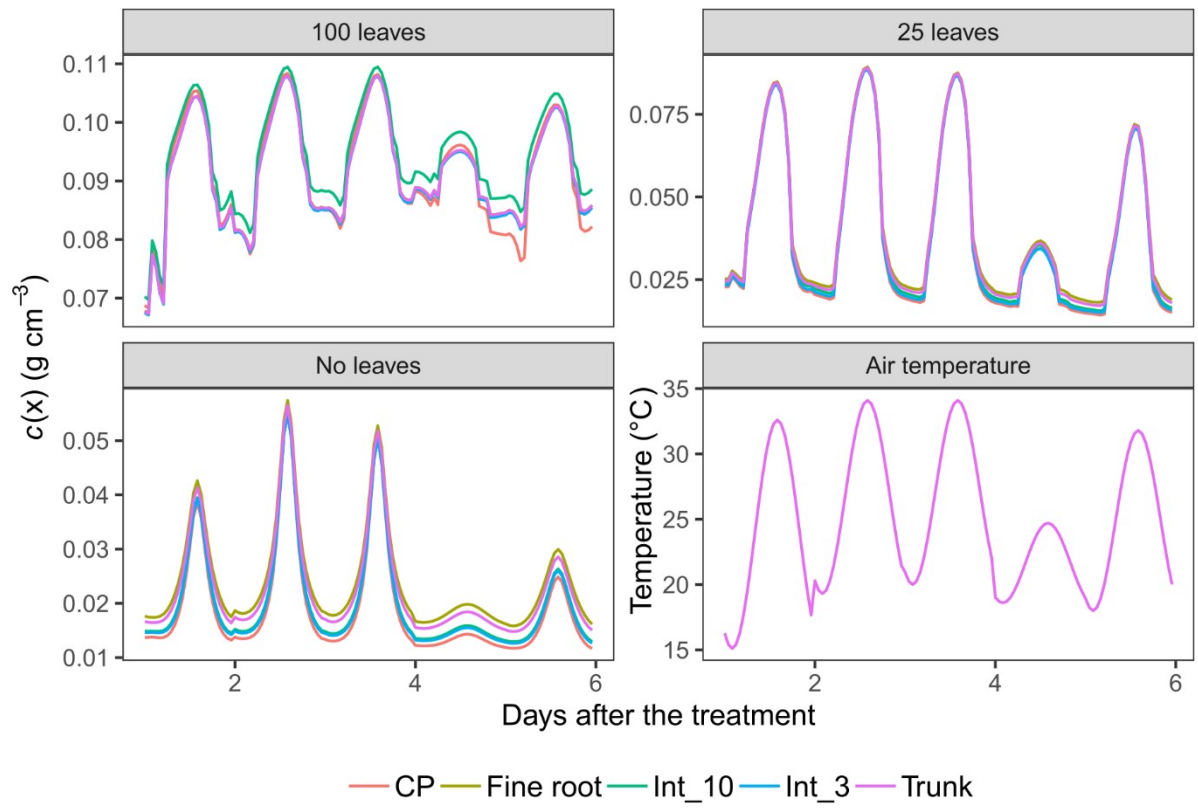


Fig. S11. The diurnal dynamics of $c(x)$ under different leaf treatments. Orange lines represent the values simulated by the common assimilate pool model, while the values at the top of internodes 10 and 3, and fine root and trunk were simulated by the coupled phloem/xylem transport model. Since the value of $c(x)$ was strongly influenced by the air temperature, the diurnal dynamics of air temperature was shown in the fourth panel.

References

- Rossouw GC, Orchard BA, Suklje K, Smith JP, Barril C, Deloire A, Holzapfel BP.** 2017. Vitis vinifera root and leaf metabolic composition during fruit maturation: implications of defoliation. *Physiol Plant* **161**, 434-450 doi: 10.1111/ppl.12604.
- Seleznyova AN, Hanan J.** 2018. Mechanistic modelling of coupled phloem/xylem transport for L-systems: combining analytical and computational methods. *Ann Bot* **121**, 991-1003 doi: 10.1093/aob/mcx204.